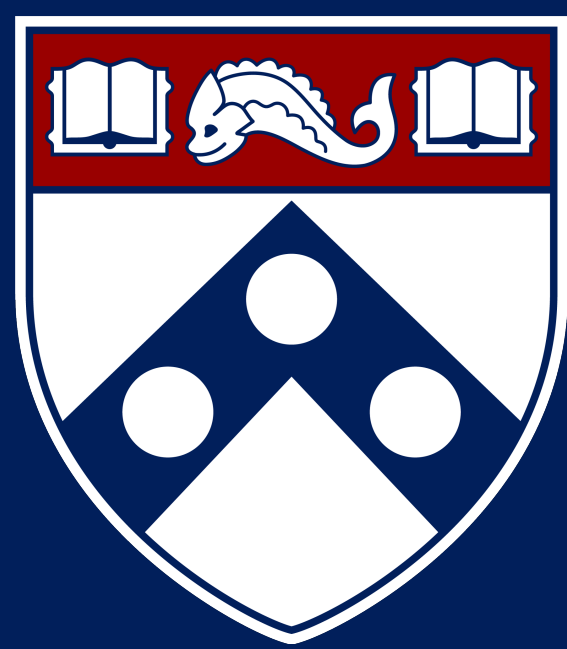


Why do marked words persist?



Corpus and experimental evidence for competition between phonological and semantic well-formedness in word-formation and the lexicon

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Takeaway

A phonologically marked form survives in a language's lexicon and is tolerated in word formation when there are fewer semantically equivalent alternative forms.

Background

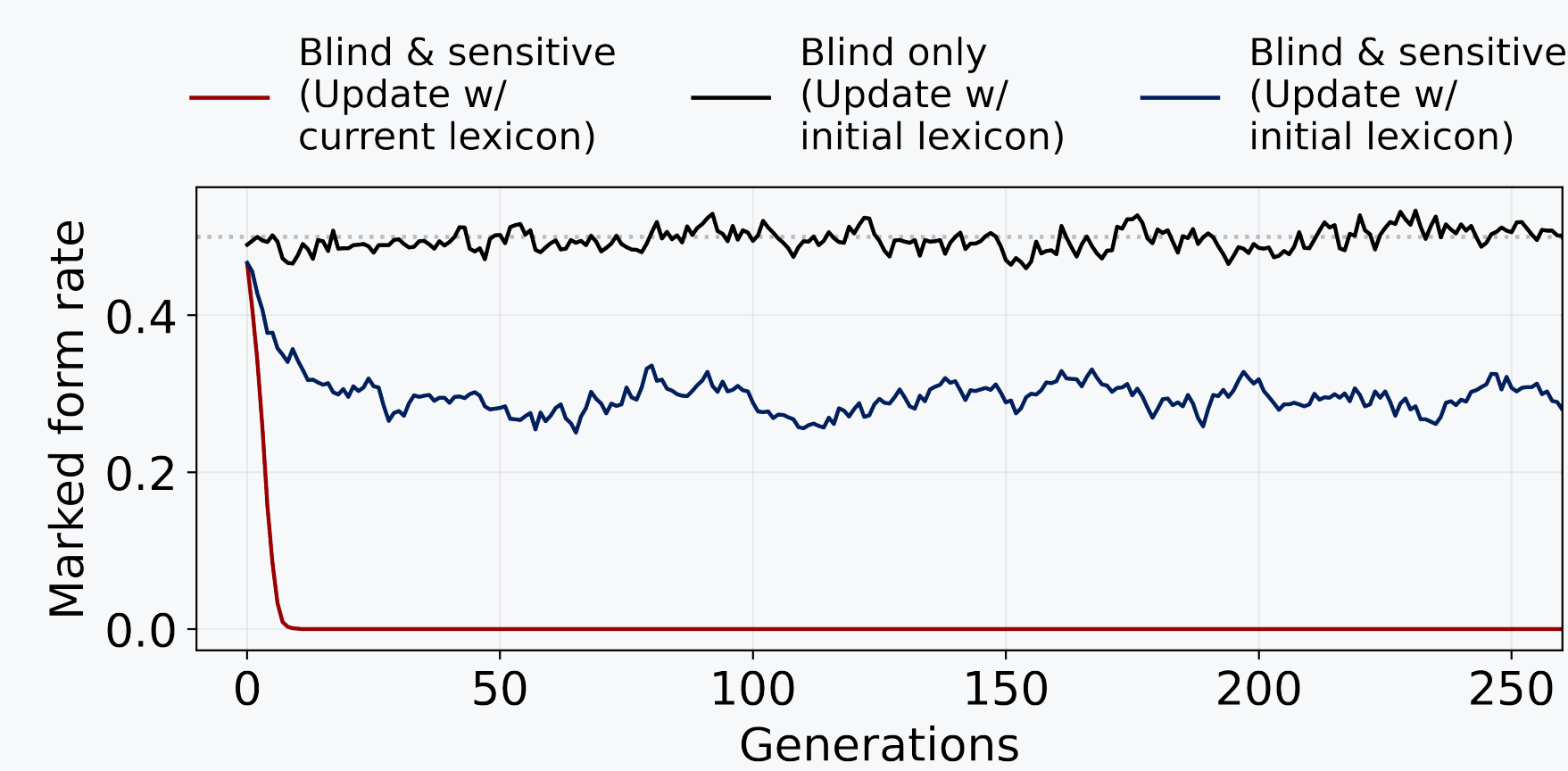
In probabilistic grammar, marked forms tend to become **marginalized** and die out over time.

Methods

MaxEnt models learned from the previous generation and produced new data for the following generation.

Table 1. Training data at initialization.

	CC	C+C
pt	100%	50%
pp	0%	50%



However, there are exceptions where marked forms are **tolerated** [1, 2, 3].

The mystery of survival: If marked forms are dispreferred, why do they survive in the lexicon?

Hypothesis

Word form flexibility (Flex) competes with phonological well-formedness

Flex= N of alternative forms compatible with the target semantics.

Corpus studies

Language	Constraint	Example
English	*GEMINATE	<i>book. keep-er</i>
Finnish	*[a _{back}][−a _{back}]	<i>kesä.loma-a</i> / 'kesæ.lomɑ:/ "summer vacation"
Hungarian	*[a _{back}][−a _{back}]	<i>kő.por</i> / 'kø:.por/ "rock flour"

Data two-member compounds from Wiktionary and published datasets [4,5]

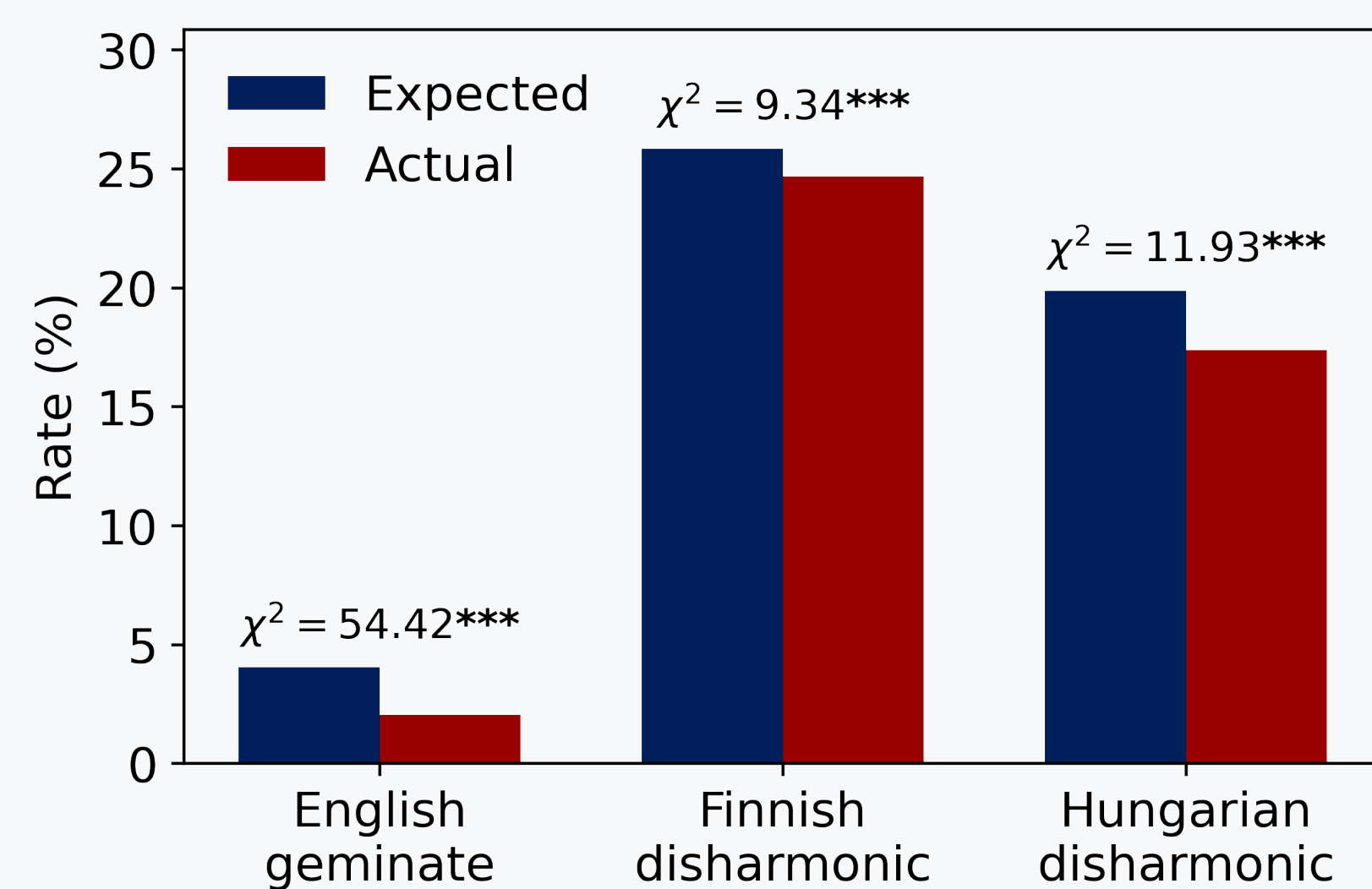
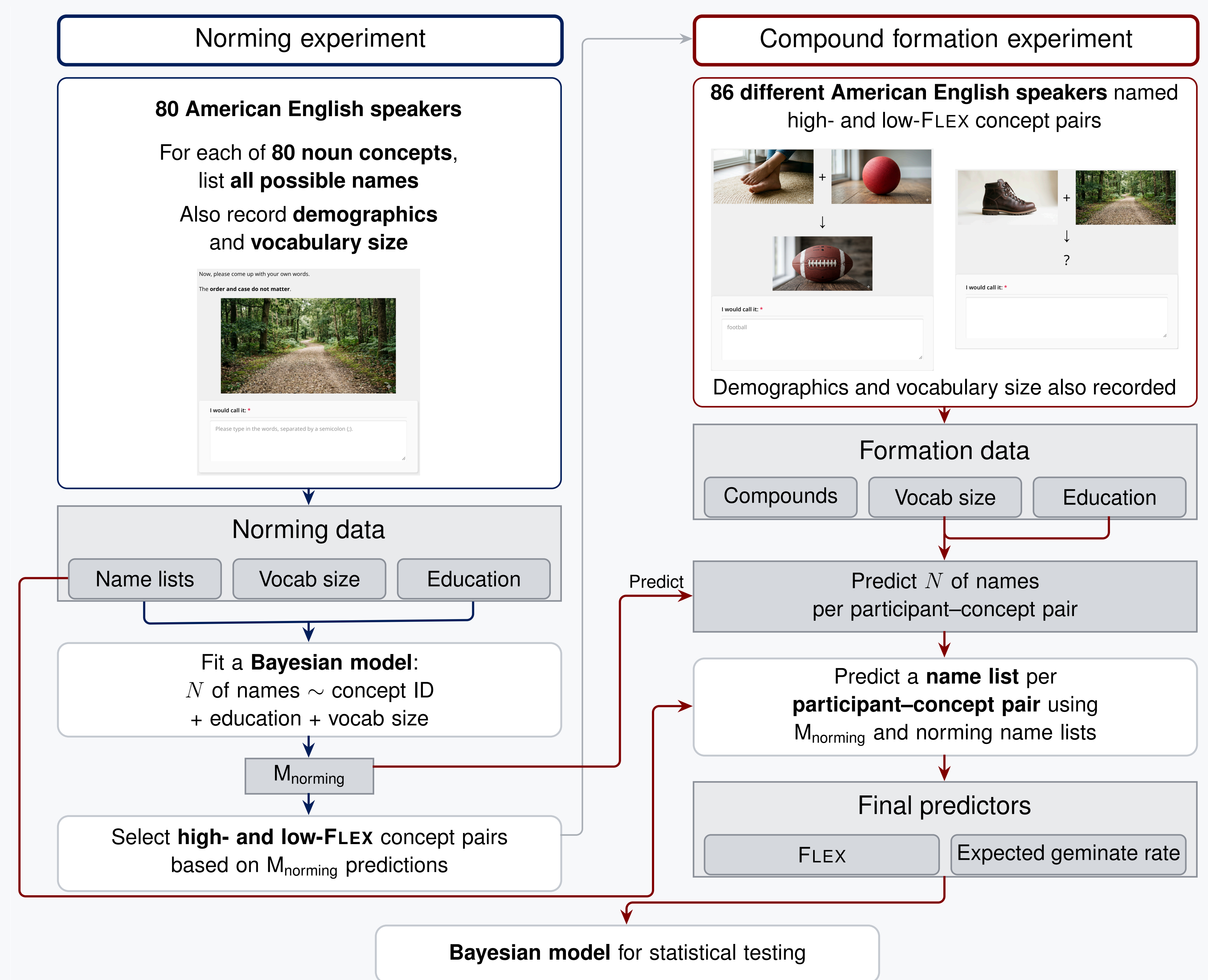


Figure 1. Geminate and disharmonic compounds are underrepresented.

Methods

- Flex(compound) = Flex(member1) × Flex(member2)
- Flex(word) = cosine similarity of the word's vector w/ all word vectors in the language

Compound formation experiment



Corpus results

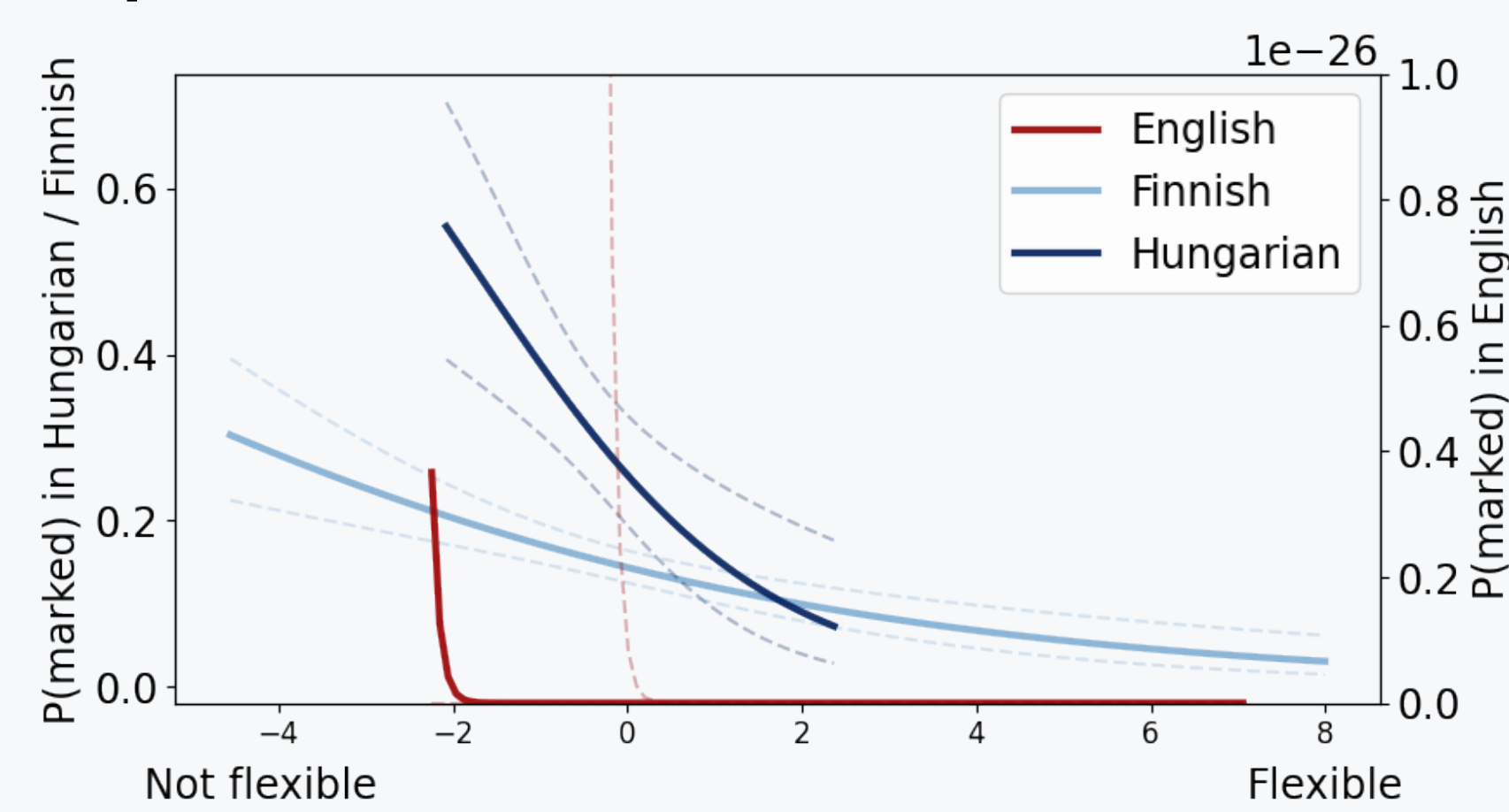


Figure 2. Flex effects on a compound's markedness.

Examining diachronic evolution through MaxEnt simulation

FLEX:

For each word:

- $k-1$ candidate forms were created, w/ the same geminate rates.
- k was randomly sampled from a Zipfian distribution (exponent= s).

Simulation results

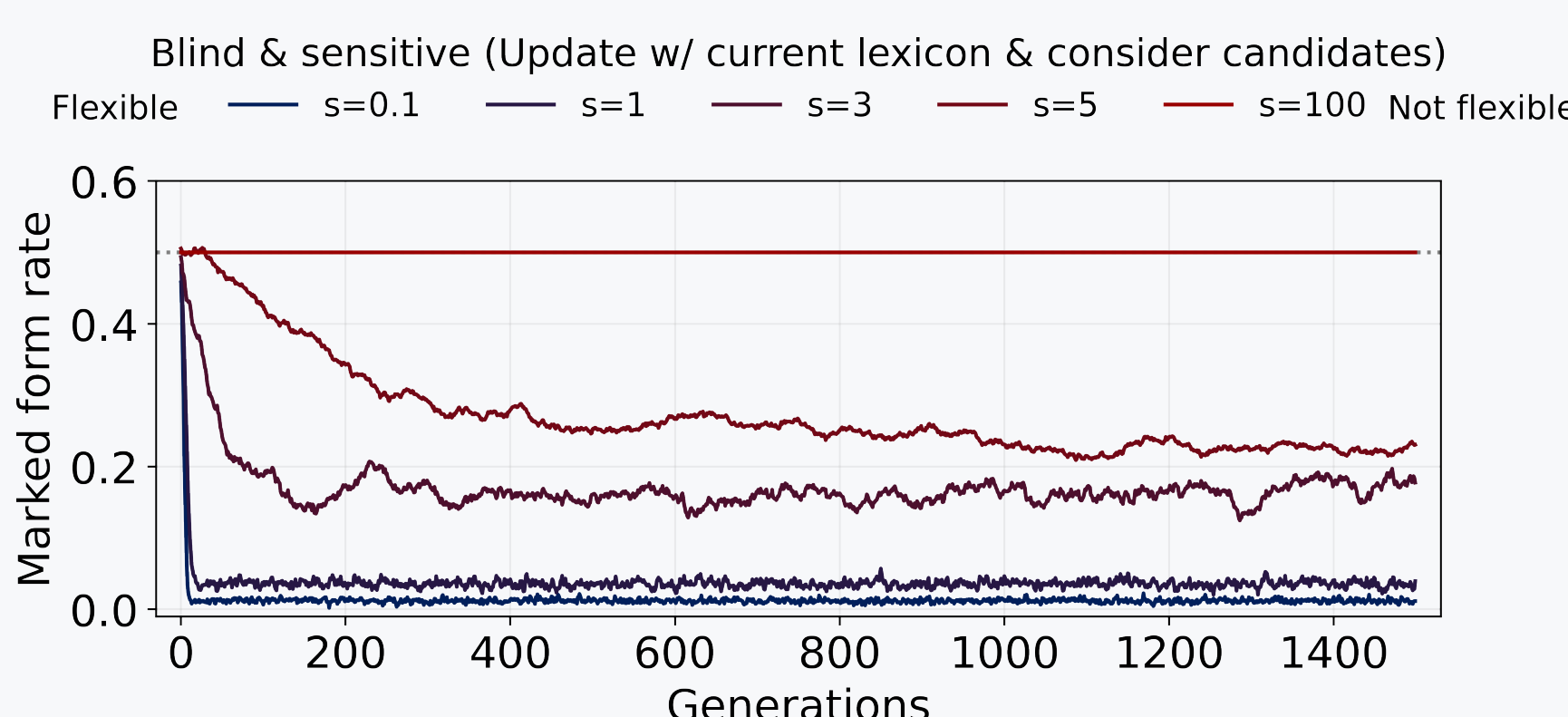


Figure 3. Simulated evolution of fake geminate compounds in the lexicon.

Experiment results

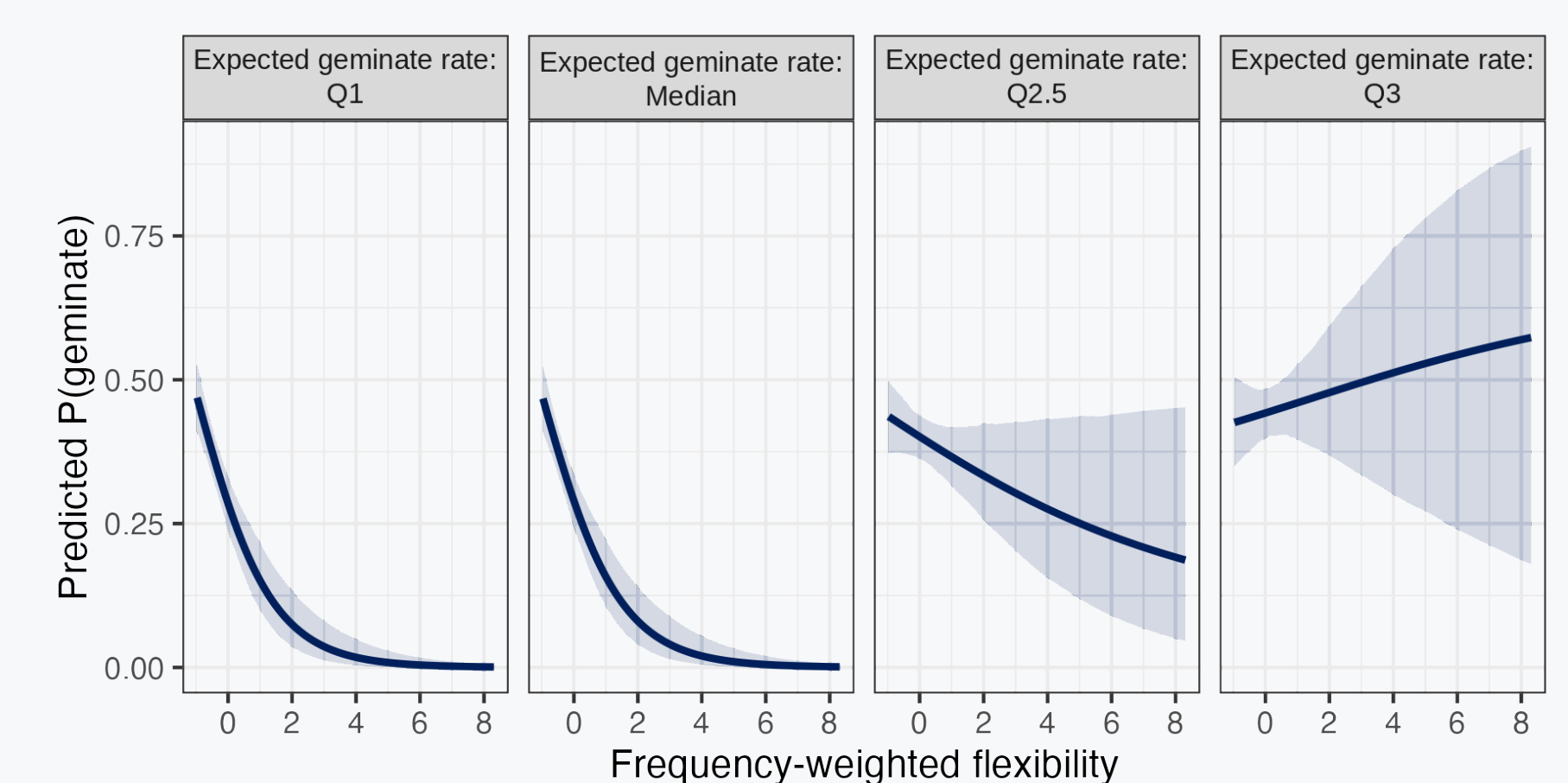


Figure 4. Model-derived interaction b/t a compound's Flex and the markedness rate of the name offered.

Selected references

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